Michael Tony

New York, NY • (551) 259-1275 • mptony6@gmail.com • linkedin.com/in/michael-tony • github.com/michaeltony6

Education

Rutgers, The State University of New Jersey – New Brunswick **September 2024 – May 2028**

B.S. Electrical Engineering, B.A. Computer Science, B.A. Mathematics *New Brunswick, NJ*

- **169 Credits Earned**
- **Electrical Engineering:** Microelectronic Processing, Computer Systems, Digital Signal Processing, Digital Electronics, Electronic Devices, Computer Architecture, Linear Systems & Signals, Circuits I & II, Digital Logic Design
- **Computer Science:** Computational Robotics, Data Science, Machine Learning, Computer Algorithms, Data Structures, MATLAB
- **Mathematics:** Calculus V (PDE), Calculus IV (ODE), Linear Algebra, Probability, Real Analysis, Mathematical Reasoning (Proofs)
- **Additional:** Physics IIA (Electricity & Magnetism) & Lab, Statics, Engineering Economics, Microeconomics, Macroeconomics, Logic, Plato & Aristotle

Work & Research Experience

Undergraduate Research Assistant – *KnowsyDev, Rutgers University* **May 2026 – Present**

- Optimized Ollama-based local LLM inference in C++/CUDA by profiling GPU kernels, memory transfers, and parallel execution, increasing token throughput by **62%**, reducing time-to-first-token by **34%**, and lowering peak VRAM consumption by **18%**.
- Built a portable interactive-document toolchain in C++, defining the IDOC grammar with Tree-sitter and compiling parsed documents into a Vulkan rendering pipeline; parsed **10,000-line files in 28 ms** with **<1 ms** incremental updates while sustaining **60 FPS** rendering.
- Developed a shared-weight multi-speaker TTS pipeline around the 100M-parameter MOSS-TTS-Nano model, benchmarking against Piper; cloned **10 voices from 10-second reference clips** and reduced deployment storage from ~**1 GB to 100 MB** (90%) while maintaining 70% speaker similarity.

Undergraduate Research Assistant – *ECE Department, Rutgers University* **May – September 2026**

- Designed and implemented the branch-predictor-inspired RowHammer detector in C++ using tagged history tables and saturating counters, **detecting 100% of evaluated attacks with zero false negatives** and a **0.3% false-positive rate**.
- Evaluated the design across **60 workloads and 18 billion memory requests**, reducing detection latency by **42%** and metadata requirements by **68%** while limiting benign-workload slowdown to **0.8%**.
- Implemented detector in SystemVerilog, synthesized it for FPGA, achieving **250 MHz**, consuming **1,500 LUTs** and **0.8% of the device's logic resources**.

Tutor – *Rutgers University Learning Center* **January 2025 – Present**

- Tutored **40+ students** through OCCM in Calculus I/II/III, Intro to CS, Data Structures, and Physics, resulting in an average **20% improvement** in grades.
- Provided mathematics and physics support to **hundreds of Rutgers student-athletes**, with tutored students maintaining a **100% course pass rate**.

Assistant Accountant – *J.D. Hayden & Co., CPAs* **July 2022 – August 2024**

- Developed the firm's client-facing website, increasing online inquiries **300%** using **JavaScript, HTML/CSS, and Google APIs**.

Project Experience

Rutgers Dynamics *(Python, C++, MuJoCo, Isaac Lab, PyTorch)*

- Developed a modular humanoid simulation/control stack using MuJoCo, MuJoCo-Warp, and Isaac Lab, integrating articulated-body dynamics, actuator models, RL environments, and GPU-parallel simulation.
- Implemented 3-DOF robotic-leg control with analytical forward/inverse kinematics, joint-space PD control, real-time feedback, and Jacobian-transpose mapping ($\tau = J^T F$) from task-space forces to actuator torques.
- Built sim-to-deployment infrastructure supporting position/velocity/torque control, torque saturation, asynchronous stepping, joint/sensor interfaces, and controller/policy validation.

Custom RISC CPU Toolchain & Pipelined Processor *(C, Assembly, Verilog, Shell)*

- Built a custom RISC CPU toolchain in C with fetch-decode-execute simulation, two-pass assembler/disassembler, fixed-width ISA encoding, tracing, profiling, and runtime fault detection.
- Designed matching Verilog CPU cores, including a full-ISA single-cycle implementation and a **five-stage pipelined core** with IF/ID/EX/MEM/WB registers, forwarding, load-use stalls, branch flushes, and CPI reporting.
- Developed automated C/Verilog regression infrastructure with golden-model state comparison, CLI breakpoints/watchpoints, and browser-based register, memory, and pipeline visualization.

Wastewater Treatment Plant ML Project *(MATLAB, Simulink, Excel)*

- Developed Gaussian Process, SVM, and decision-tree models in MATLAB to predict wastewater effluent properties, **improving prediction accuracy 30%**.
- Preprocessed **350K+ data points** through randomization and normalization, **reducing SVM prediction error by 80%+** and mitigating overfitting.

Unilert | 1st Place – Dedalus Track, HackPrinceton *(Python, FastAPI, TypeScript, Next.js, Dedalus Machines, K2 Think V2, OSRM)*

- Built a real-time EMS/ICS decision-support system that transforms incident reports into patient routing, bottleneck alerts, and role-owned actions.
- Architected a fault-tolerant deterministic/agentic pipeline with four K2 Think V2 agents on Dedalus Machines; integrated NWS/USGS hazard data, routing, and hospital metadata with **38/38 automated tests passing**.

Skills

- **Languages:** C, C++, Python, Java, MATLAB, SystemVerilog/Verilog, Assembly (RISC-V, x86, x64), JavaScript/TypeScript, SQL, Shell, LaTeX
- **ML/Robotics:** PyTorch, TensorFlow, NumPy, MuJoCo, MuJoCo-Warp, Isaac Lab, Simulink, RSL-RL, Gaussian Processes, SVMs, Decision Trees
- **Systems/Tools:** CUDA, Vulkan, Linux, Git, GitHub Actions, CMake, Tree-sitter, Xilinx Vivado, Icarus Verilog, KiCad, FastAPI, Next.js, SolidWorks, Arduino